SmartGrid-ES V2 Results Report

Reinforcement Learning, Wrappers, and Results Analysis

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Report purpose. This report presents the V2 experimental results of SmartGrid-ES, a reinforcement learning project focused on simplified energy management in a discrete smart-grid environment. It summarizes the theoretical basis, execution workflow, numerical outputs, and interpretation of results for the reported scenarios.

Abstract

This report presents the V2 results of SmartGrid-ES, an academic reinforcement learning project based on a small discrete environment and a tabular Q-learning agent. The objective is to model a simplified smart-grid decision problem in which the agent must decide how to manage battery usage, renewable generation, and electricity demand. Compared with the initial version, V2 introduces scenario-based environment construction, wrapper-based behavior changes, comparative evaluation, and result artifact generation for analysis. The report focuses on the `baseline` and `combined_v2` scenarios, explaining the main aggregate metrics, episode-level behavior, and the most relevant limitations identified during interpretation.

Keywords: reinforcement learning, Q-learning, Gymnasium, wrappers, smart grid, energy management, results analysis.

Contents

1 Introduction 2
2 Theoretical Context 2
2.1 Reinforcement Learning and Q-learning . . . 2
2.2 MDP Formulation . . . 2
2.3 Gymnasium and Wrappers . . . 2
3 Project Description 3
3.1 Environment Design . . . 3
3.2 Selected Scenarios . . . 3
4 Experimental Setup 3
4.1 Workflow . . . 3
4.2 Configuration . . . 4
4.3 Result Provenance . . . 4
5 Metric Definitions and Reading Guide 4
5.1 Aggregate Metrics . . . 4
5.2 Episode Variables . . . 5
5.3 How to Read the Charts . . . 5
6 Results 6
6.1 Main Findings . . . 6
6.2 Comparative Evaluation Summary . . . 6
6.3 Aggregate Comparison Charts . . . 7
6.4 Operational Interpretation of the Two Scenarios . . . 7
6.5 Episode-Level Tables . . . 8
6.6 Baseline Episode Charts . . . 9
6.7 Combined_v2 Episode Charts . . . 10
6.8 Interpretation of Episode-Level Behavior . . . 10
7 Discussion 11
7.1 Main Interpretation . . . 11
7.2 Strength of the V2 Design . . . 11
7.3 Metric Caveat . . . 11
8 Limitations and Future Work 12
9 Conclusion 12

1 Introduction

Artificial Intelligence includes methods that perceive, reason, learn, and act in order to solve problems and adapt to new situations. Within this broader field, reinforcement learning is especially useful when an agent must improve its behavior through repeated interaction with an environment. In this setting, the agent observes a state, selects an action, receives a reward, and gradually learns a policy that maximizes long-term return.

The SmartGrid-ES project applies this idea to a simplified energy-management problem inspired by smart-grid operation. Instead of attempting to simulate a full electrical grid, the project uses a compact and discrete environment that is easier to understand, defend, and extend in an academic setting. This design choice keeps the system manageable while preserving the essential logic of sequential decision making in an energy domain.

The report focuses on two representative scenarios:

- **baseline**, which serves as the reference case;
- **combined_v2**, which introduces a more demanding operating configuration by combining several V2 modifications.

2 Theoretical Context

2.1 Reinforcement Learning and Q-learning

In reinforcement learning, an agent learns from interaction rather than from explicit labeled examples. The core loop is simple: the agent chooses an action, the environment responds with a reward and a new state, and the agent updates its behavior accordingly.

The SmartGrid-ES project uses **Q-learning**, a classical tabular reinforcement learning algorithm. Q-learning estimates a value $Q(s, a)$ for each state–action pair and updates that value according to the Bellman equation:

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha \left(R + \gamma \max_{a'} Q(s', a') \right)$$

This approach is appropriate here because the environment is intentionally discrete, so the state–action space can be represented through a compact Q-table.

2.2 MDP Formulation

The project can be interpreted through the standard Markov Decision Process formalism:

$$(S, A, P, R, \gamma)$$

where S is the set of states, A is the set of actions, P is the transition dynamics, R is the reward function, and γ is the discount factor.

This perspective is useful because it connects the implementation to the theoretical framework of reinforcement learning. The agent does not need the full history of previous observations if the current state contains the relevant information for decision making.

2.3 Gymnasium and Wrappers

Gymnasium provides a standardized API for reinforcement learning experiments. Its interaction loop is based on environment creation, `reset()` to start an episode, and `step(action)` to advance the environment while returning observation, reward, termination status, truncation status, and auxiliary information.

A major conceptual contribution of V2 is the use of **wrappers**. Wrappers modify an existing environment without rewriting the base code. This keeps the design modular and makes scenario-based experimentation substantially cleaner. In SmartGrid-ES V2, wrappers are used to alter seasonal behavior, demand variability, battery losses, and reward shaping.

3 Project Description

3.1 Environment Design

The environment models a simplified microgrid using four discrete state variables:

- battery level,
- demand level,
- renewable generation level,
- period of the day.

The agent chooses among four discrete actions.

Table 1: Action encoding used in the environment

Action ID	Meaning
0	use battery
1	buy from grid
2	store surplus energy
3	sell surplus energy

This abstraction is intentionally small, but it still captures the main trade-offs of the problem:

- meeting demand,
- minimizing dependence on the external grid,
- using battery storage efficiently,
- exploiting renewable surplus when possible.

3.2 Selected Scenarios

The present report concentrates on:

- **baseline**: the reference environment without additional complexity;
- **combined_v2**: a more challenging scenario that combines several wrapper-based effects.

These two scenarios provide a compact but meaningful V2 comparison. The first one shows the behavior of the core system under reference conditions, while the second stresses the policy under a richer and more demanding operating profile.

4 Experimental Setup

4.1 Workflow

The execution workflow followed these steps:

1. train the Q-learning agent on the baseline scenario;
2. train the Q-learning agent on the combined_v2 scenario;
3. run the evaluation script on the available trained models;
4. inspect the generated artifacts and summarize the results.

The commands used were:

Listing 1: Execution commands used for the reported V2 run

```
python -m src.training.train --scenario baseline
python -m src.training.train --scenario combined_v2
python -m src.training.evaluate
```

4.2 Configuration

The main experimental parameters were:

- random seed: 42,
- baseline training episodes: 2000,
- combined_v2 training episodes: 4000,
- maximum steps per episode: 24,
- evaluation episodes: 100.

These settings are sufficient for a clean V2 academic demonstration: they keep the pipeline lightweight while still producing meaningful artifacts for comparison.

4.3 Result Provenance

All numerical values reported below are aligned with the artifacts committed in the `v2-main` branch. The aggregate metrics come from the evaluation summary file, while the episode-level charts come from the demo CSV files associated with `baseline` and `combined_v2`. This guarantees that the narrative of the report is synchronized with the repository outputs.

5 Metric Definitions and Reading Guide

This section explains the meaning of the reported variables and how the charts should be interpreted.

5.1 Aggregate Metrics

Table 2 summarizes the interpretation of the evaluation metrics used in this report.

Table 2: Meaning of the main aggregate metrics

Metric	Interpretation
`avg_reward`	Mean cumulative reward obtained during evaluation episodes. Higher values are better only with respect to the implemented reward design.
`avg_coverage`	Mean demand coverage ratio. In this run, one scenario produced a value above 1.0, so this metric must be interpreted with caution.
`avg_grid_bought`	Mean amount of energy purchased from the external grid. Higher values indicate stronger dependence on outside supply.
`avg_battery_end`	Mean battery level at the end of an episode. Higher values indicate that the policy tends to preserve more stored energy.
`avg_sold`	Mean amount of energy sold as surplus. Higher values indicate more opportunities to export unused energy.

5.2 Episode Variables

The episode-level charts use the following variables:

- `battery`: discrete battery level available at each step;
- `demand`: discrete demand level observed at the step;
- `renewable`: discrete renewable generation level observed at the step;
- `action`: action selected by the agent at that step;
- `reward`: immediate reward returned by the environment after the action;
- `grid_bought`: energy bought from the external grid at the step;
- `demand_covered`: amount of demand satisfied at the step;
- `unmet_demand`: demand that remained uncovered.

5.3 How to Read the Charts

- In the aggregate bar charts, the x-axis identifies the scenario or the metric being compared, and the y-axis represents the numerical value of the metric.
- In the episode line charts, the x-axis represents the decision step, from 0 to 23.
- In the state-evolution plots, the y-axis represents the discrete level of each variable.
- In the reward plots, the y-axis represents the immediate reward received at each step.
- In the action-distribution charts, the x-axis represents the action identifier and the y-axis represents the frequency with which that action appears in the selected demo episode.

6 Results

6.1 Main Findings

Main findings

- `combined_v2` achieved a higher average reward than `baseline`.
- `combined_v2` also required substantially more electricity purchases from the external grid.
- The final battery level was much lower in `combined_v2`, which suggests a more constrained operating profile.
- The `avg_sold` metric also decreased sharply in `combined_v2`, indicating fewer opportunities to preserve and export surplus energy.
- The higher reward should therefore not be interpreted as an unconditional improvement in every operational dimension.

6.2 Comparative Evaluation Summary

The main evaluation summary is reproduced in Table 3.

Table 3: Evaluation summary for the two reported scenarios

`Scenario`	Avg. reward	Avg. coverage	Avg. grid bought	Avg. battery end	Avg. sold
`baseline`	42.420	1.0000	5.93	0.81	4.62
`combined_v2`	62.871	1.0699	14.26	0.03	0.68

The summary already shows a clear trade-off: the more demanding scenario reaches a higher reward, but it does so while relying more heavily on external grid purchases, ending with almost no stored battery, and selling substantially less surplus energy.

6.3 Aggregate Comparison Charts

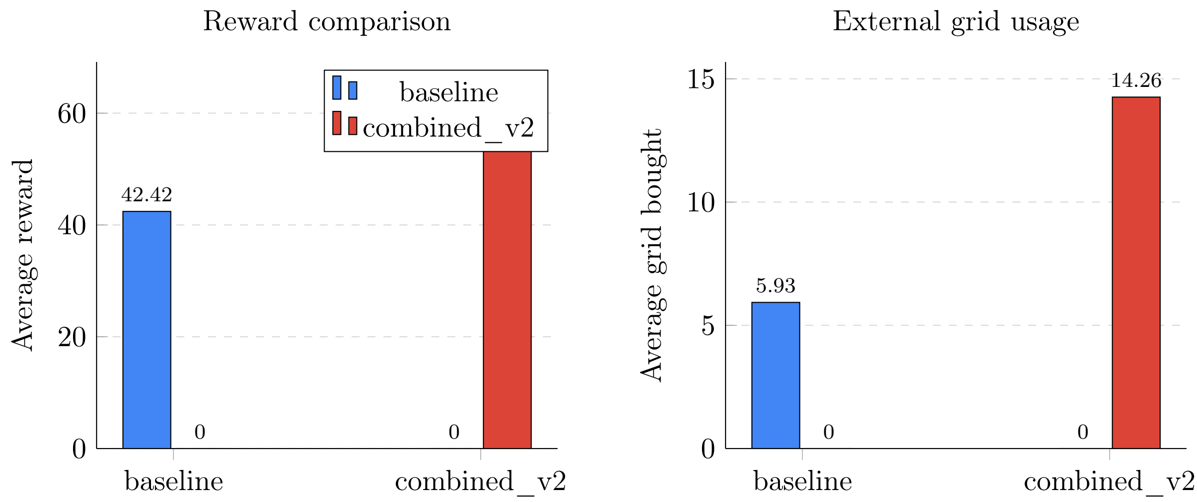

Figure 1: Comparison of average reward and average external grid purchases. In both panels, the two scenarios are distinguished by color. The charts show that `combined_v2` reaches a higher reward, but this is accompanied by a much larger amount of energy bought from the external grid.

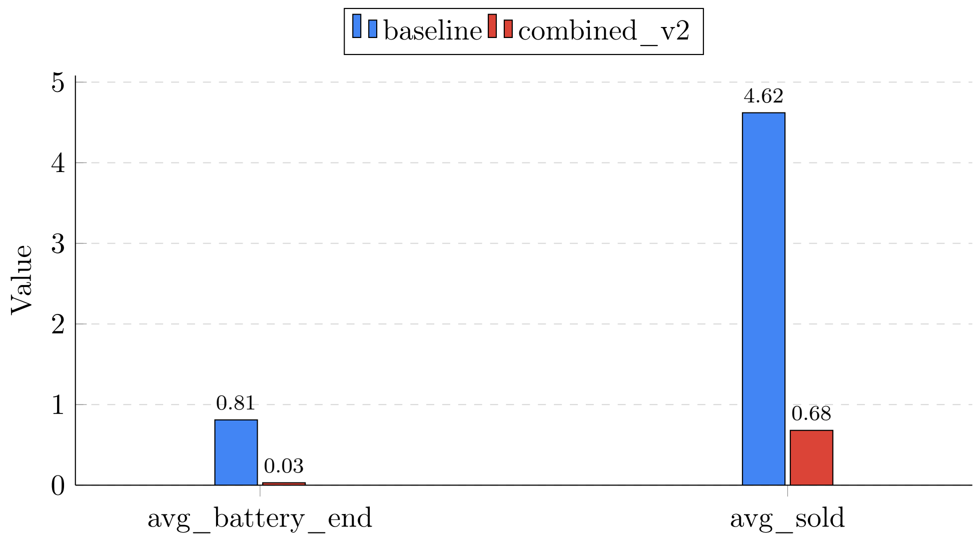

Figure 2: Comparison of average final battery level and average sold energy. The chart shows that the more demanding scenario ends with almost no battery reserve and much lower sold energy, which suggests a policy operating under tighter or more stressed conditions.

6.4 Operational Interpretation of the Two Scenarios

Taken together, Figures 1 and 2 support an important conclusion: reward alone is not enough to describe performance. The `combined_v2` scenario obtains a higher average reward, but it does so while buying more from the grid, preserving almost no final battery reserve, and selling much less surplus energy.

This suggests that the agent adapts to a more difficult scenario by adopting a policy that is still successful according to the reward function, but more dependent on outside energy and less able to maintain an efficient storage profile.

6.5 Episode-Level Tables

To complement the aggregate comparison, the repository also provides step-by-step demo episodes for both reported scenarios. Tables 4 and 5 show representative excerpts chosen to illustrate different operating moments, including high-demand states, low-renewable states, battery transitions, and reward penalties.

Table 4: Excerpt from the baseline demo episode

Step	Battery	Demand	Renewable	Action	Reward
0	0	2	2	3	3.00
1	0	2	2	0	3.00
5	1	2	0	1	2.80
9	0	2	1	1	3.40
14	2	1	1	2	1.00
19	2	1	0	1	1.40
22	2	2	0	0	4.00
23	0	0	0	3	-1.00

Table 5: Excerpt from the combined_v2 demo episode

Step	Battery	Demand	Renewable	Action	Reward
0	0	2	2	0	3.25
5	0	2	2	1	3.30
8	0	2	1	1	3.90
12	0	0	1	0	-0.75
18	0	1	0	3	2.90
20	0	1	0	1	3.90
22	0	1	0	1	3.30
23	0	2	2	1	1.90

6.6 Baseline Episode Charts

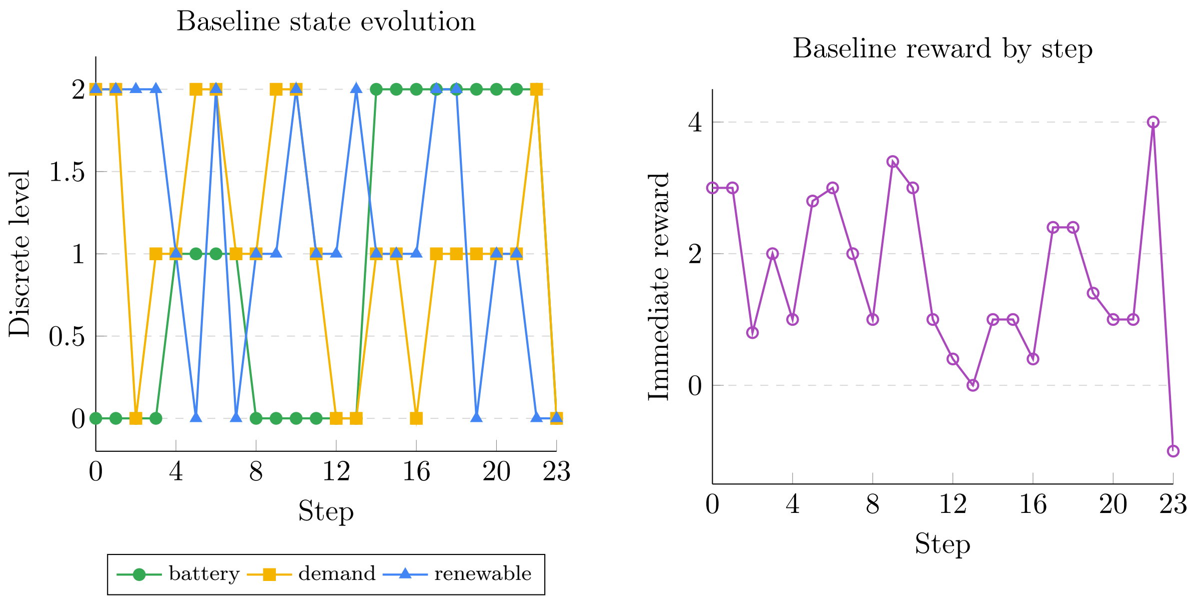

Figure 3: Baseline episode behavior. The left panel shows the evolution of battery, demand, and renewable generation across the 24-step episode. The right panel shows the immediate reward returned at each step. Together, both panels indicate a non-trivial policy reacting to changing operating conditions rather than following a static pattern.

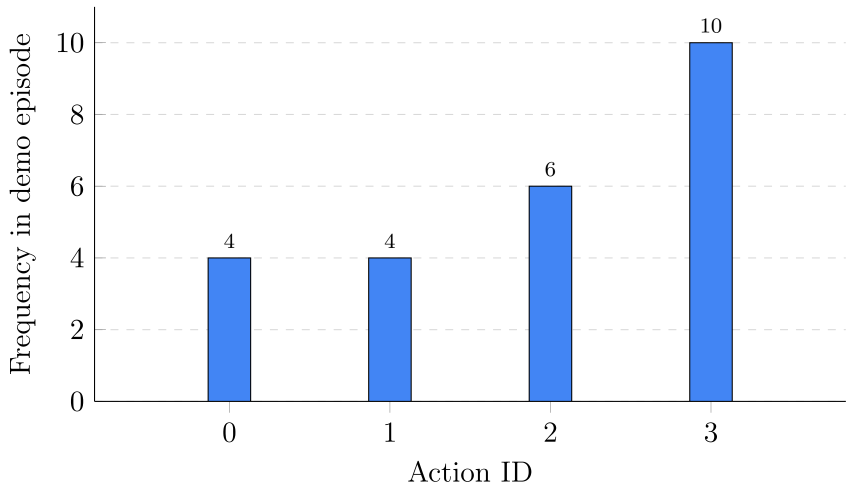

Figure 4: Baseline action distribution. Action 3 is the most frequent in this episode, followed by action 2, which indicates that the policy is not dominated by a single fixed response pattern.

6.7 Combined_v2 Episode Charts

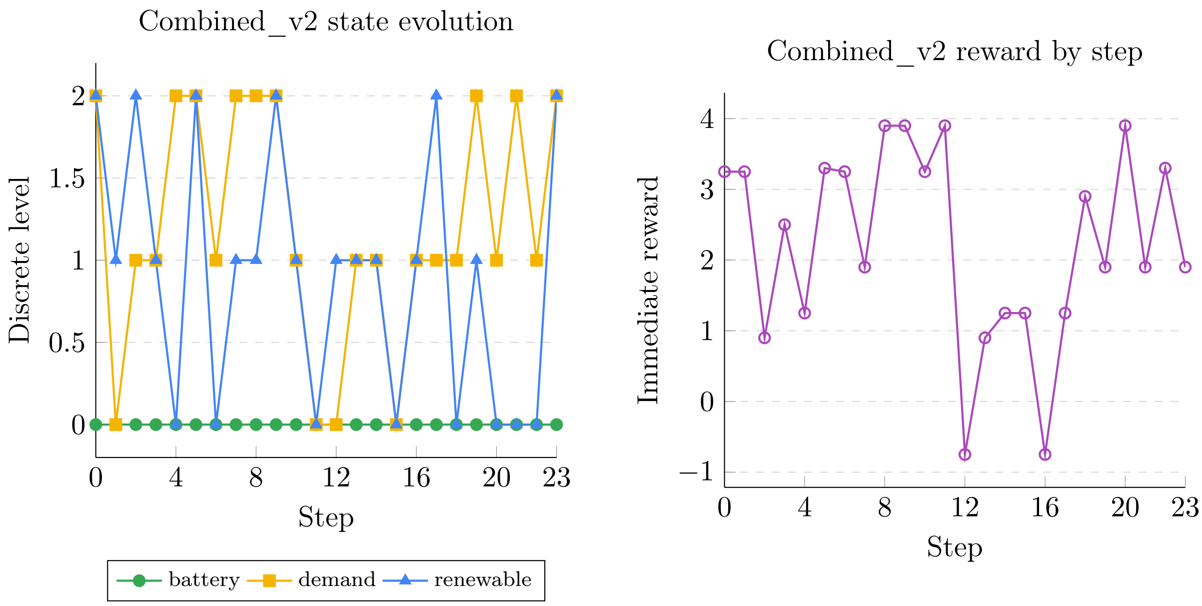

Figure 5: Combined_v2 episode behavior. The left panel shows battery, demand, and renewable generation over the 24-step episode. The right panel shows the immediate reward sequence. Unlike the baseline episode, the battery remains at zero throughout the entire episode, which is consistent with the very low average final battery level reported in the aggregate results.

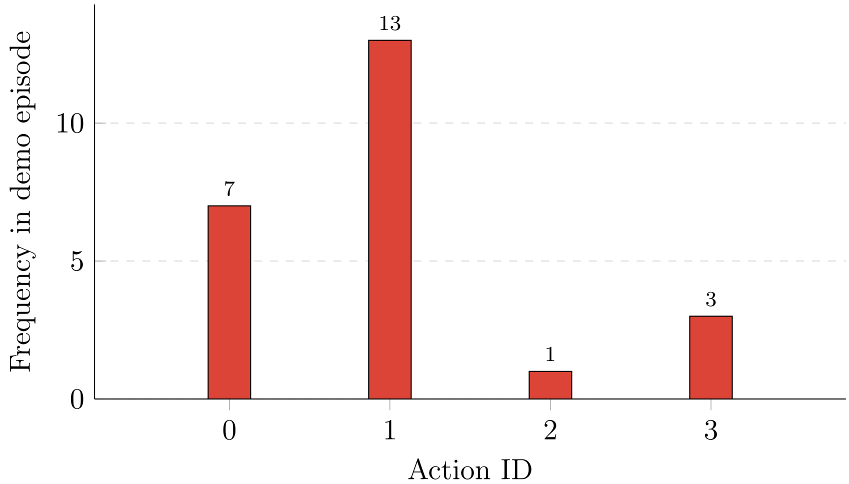

Figure 6: Combined_v2 action distribution. Action 1 dominates the episode, suggesting that the scenario pushes the agent toward a more repetitive and externally supported control strategy.

6.8 Interpretation of Episode-Level Behavior

The episode-level plots are useful because they show how the agent behaves operationally and not only through aggregate statistics. In the baseline episode, the battery evolves across several discrete levels and the action histogram is more distributed. In the combined scenario, the battery remains at zero throughout the reported episode and the action distribution becomes much more concentrated around action 1.

This difference matters because it provides a concrete explanation for the aggregate results. The higher `avg_grid_bought`, the lower `avg_battery_end`, and the lower `avg_sold` reported for `combined_v2` are consistent with the sequential behavior observed in the demo episode.

7 Discussion

7.1 Main Interpretation

The results suggest that the V2 architecture is already useful for comparative experimentation. The baseline scenario provides a stable reference case, while `combined_v2` stresses the policy under more complex operating conditions.

The higher reward of `combined_v2` should not be interpreted in isolation. It appears together with higher grid purchases, a much lower remaining battery level, and far less sold surplus energy. Therefore, the richer scenario seems to push the agent toward a different operating strategy rather than simply producing a uniformly better policy in every dimension.

From an energy-management point of view, the main conclusion is not that `combined_v2` is better, but that it reveals a policy adapted to a more constrained scenario, with a clear trade-off between reward maximization and operational efficiency.

7.2 Strength of the V2 Design

One of the main strengths of V2 is architectural rather than purely numerical. The project now includes:

- scenario-based environment construction,
- wrapper-based modular extensions,
- generated CSV summaries,
- reproducible result artifacts,
- a results folder that supports verification and traceability.

This is important because V2 is no longer just a minimal reinforcement learning prototype. It is now a compact experimentation pipeline that supports explanation, comparison, and future extension.

7.3 Metric Caveat

One issue must be stated explicitly: the reported `avg_coverage` value for `combined_v2` is 1.0699, which is above 1.0. If coverage is interpreted as the fraction of demand that was satisfied, then this quantity would usually be expected to remain in the interval $[0, 1]$.

This does not invalidate the execution itself, but it strongly suggests one of the following:

- the current coverage metric is not clipped,
- the underlying `demand_covered` variable can exceed the instantaneous demand in some transitions,
- or the implemented metric is capturing a slightly different quantity than normalized demand coverage.

For that reason, the coverage result should be treated as informative but not yet perfectly calibrated.

8 Limitations and Future Work

The present V2 report is intentionally limited to a small and controlled setup. The main limitations are the following:

- only two scenarios are documented in detail here,
- the environment remains deliberately simplified and discrete,
- the project uses tabular Q-learning rather than function approximation,
- at least one reported metric requires further validation.

These limitations are acceptable for a V2 academic milestone, but they also suggest several clear next steps:

1. validate and, if necessary, correct the coverage metric,
2. evaluate all available scenarios under the same reporting pipeline,
3. enrich the environment dynamics,
4. incorporate more realistic energy assumptions or external data,
5. explore future extensions toward multi-agent or richer smart-grid formulations.

9 Conclusion

SmartGrid-ES V2 moves beyond a minimal reinforcement learning prototype and becomes a more structured experimental system. The combination of a discrete environment, tabular Q-learning, modular wrappers, scenario-based execution, stored result artifacts, and a clear evaluation pipeline makes the project easier to analyze and defend.

The comparison between `baseline` and `combined_v2` shows that the agent behaves differently under more demanding conditions, and that reward alone is not enough to describe performance. Grid dependence, battery usage, and sold surplus energy also matter for interpretation.

Overall, V2 can be considered a solid intermediate milestone: it remains small enough to be understandable, yet rich enough to demonstrate progression, experimentation, and meaningful analysis.